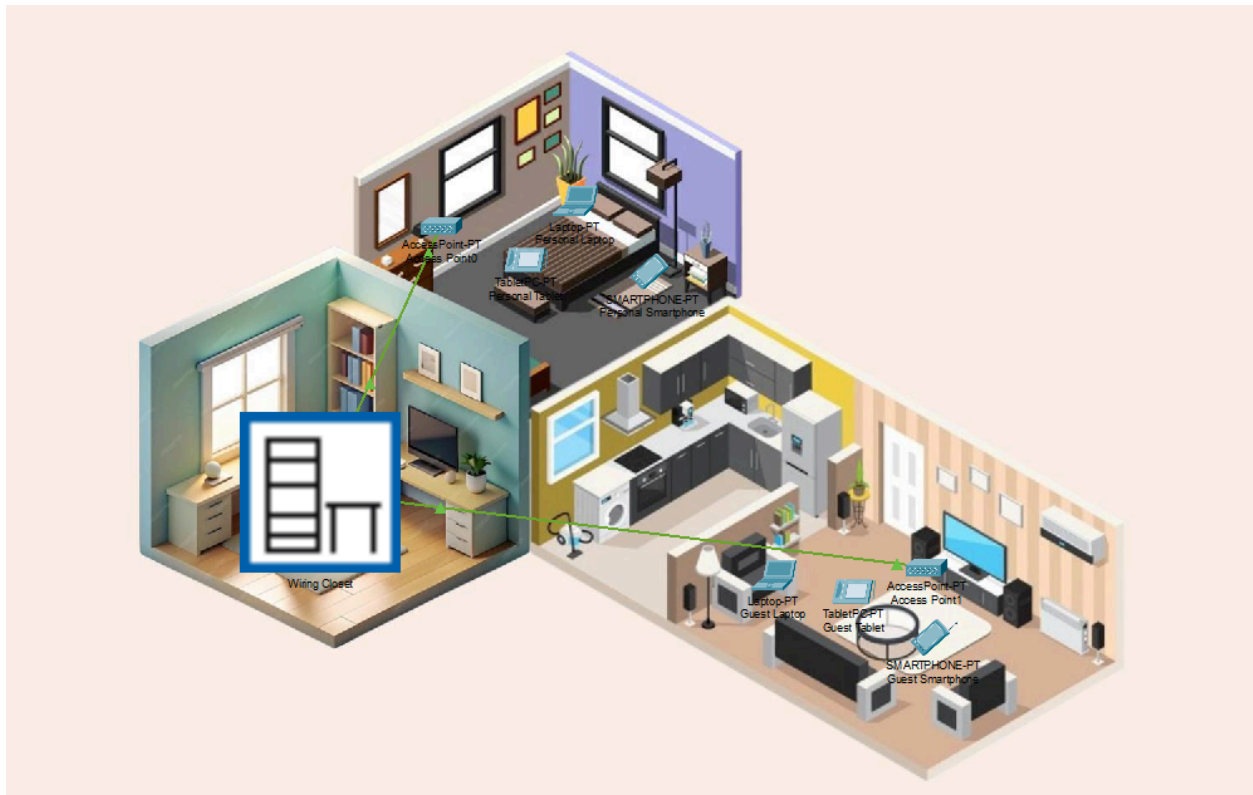


Low-Level Documentation for Home Office Network Configuration

This document outlines the physical setup of a home office network using Cisco Packet Tracer.



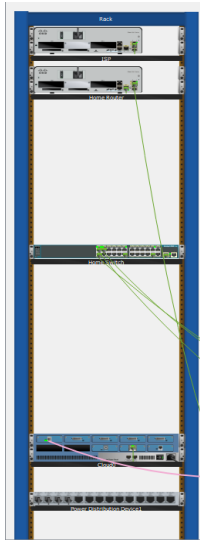
Assumptions and Constraints

- **Assumptions:**
 - All required network devices are available in Cisco Packet Tracer.
 - The network will be implemented in a home office environment, simulating a small-scale network with multiple VLANs.
 - The wiring closet in the office of the home serves as the home office of the client.
- **Constraints:**
 - The power supply must support all network devices without interruption.

- The network simulation assumes a stable ISP connection via the WAN emulation cloud (Cloud0) for accurate testing and performance evaluation.

Diagrams of Rack, Table, and Slot Layouts

Rack Layout Overview



Top Shelf:

- **ISP Router (Cisco 1941):**
 - This router simulates the ISP connection, acting as the gateway to the internet.
- **Home Router (Cisco 1941):**
 - Positioned on the top shelf to serve as the central gateway for the network.
 - Manages inter-VLAN routing and connections to the ISP.
 - This location minimizes cable clutter, offering easy access for connections to other networking devices.

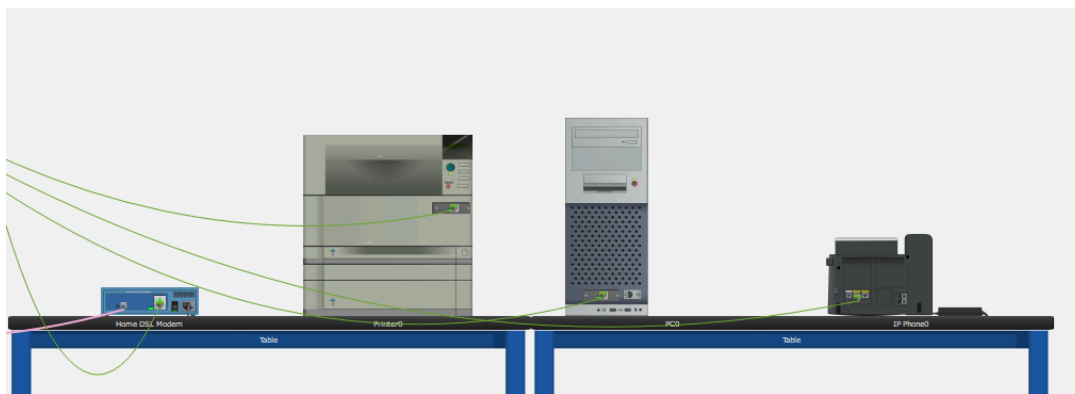
Middle Shelf:

- **Home Switch (Cisco 2960):**
 - Centrally located to efficiently manage connections to devices across different VLANs.

Bottom Shelf:

- **Cloud Device (Cloud0):**
 - Positioned on the bottom shelf to maintain organization and reduce heat buildup.
- **Power Distribution Device:**
 - Placed on the bottom shelf due to its weight, lowering the center of gravity and stabilizing the rack.
 - Allows easy access to power outlets, facilitating efficient cable management.

Table Layout Overview



- **Printer:** Positioned to the left side of the table for easy access when printing documents.
- **PC:** Centrally located on the table, serving as the primary workstation.
- **DSL Modem:** Placed close to the power outlet, and connected to the router for internet access.
- **Printer:** Positioned to the right side of the table, within reach for quick communication.

Slot Layouts



ISP and Home router

GigabitEthernet 0/0 (ISP Router):

- **Connection:** Cloud0.

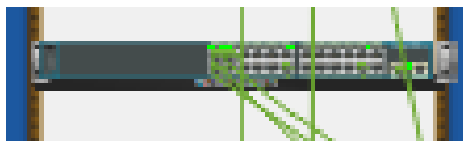
- **Purpose:** Provides simulated internet connectivity by linking the ISP router to the WAN emulation cloud.

GigabitEthernet 0/0 (Home Router):

- **Connection:** ISP via Cloud0.
- **Purpose:** Provides internet connectivity by linking the home router to the ISP router through the cloud simulation.

GigabitEthernet 0/1 (Home Router):

- **Connection:** Home Switch.
- **Purpose:** Handles internal network traffic by connecting the router to the home switch.



Switch



Access Points

FastEthernet 0/1-10 (Switch):

- **Assigned to:** VLAN 10 (Work Devices).
- **Purpose:** Connects work-related devices to the network.

FastEthernet 0/11-20 (Switch):

- **Assigned to:** VLAN 20 (Personal Devices).
- **Purpose:** Connects personal devices to the network.
- **Connected Device:** Access Point (for Personal Devices) FastEthernet 0/11 (Switch).

FastEthernet 0/21-24 (Switch):

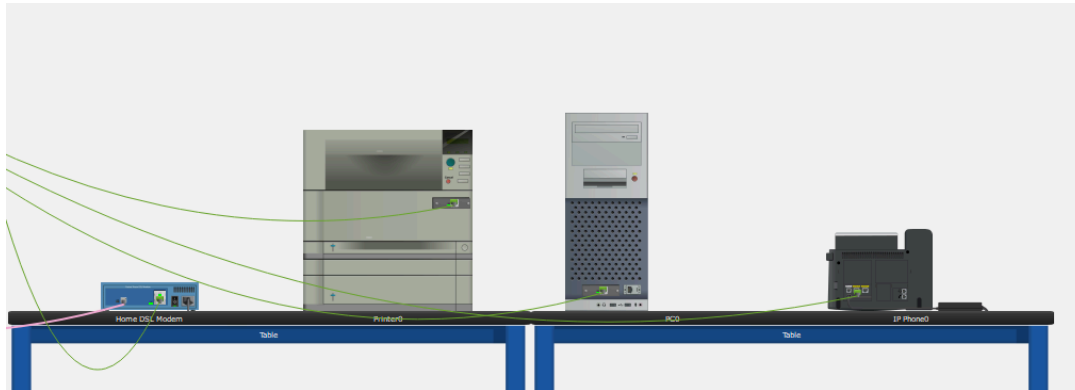
- **Assigned to:** VLAN 30 (Guest Devices).
- **Purpose:** Connects guest devices to the network.
- **Connected Device:** Access Point (for Guest Devices) FastEthernet 0/22 (Switch).

GigabitEthernet 0/1 (Switch):

- **Connection:** Trunk to Home Router.
- **Purpose:** Carries traffic for multiple VLANs between the switch and the router.

End Device Layouts

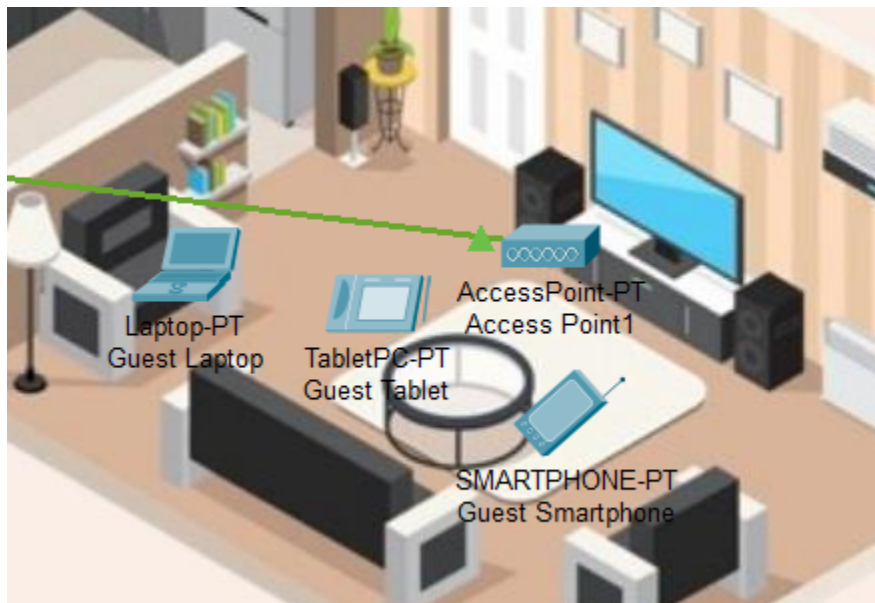
Wired Connections



1. **PC (Work Device)**
 - **VLAN:** 10 (Work Devices)
 - **Connection:** FastEthernet 0/1 (Switch)
 - **Location:** On the table
2. **Printer (Work Device)**
 - **VLAN:** 10 (Work Devices)
 - **Connection:** FastEthernet 0/2 (Switch)
 - **Location:** On the table
3. **Phone (Work Device)**
 - **VLAN:** 10 (Work Devices)
 - **Connection:** FastEthernet 0/3 (Switch)
 - **Location:** On the table
4. **DSL Modem**
 - **VLAN:** Not assigned (Connected directly to ISP)
 - **Connection:** GigabitEthernet 0/0 (Home Router)
 - **Location:** On the table

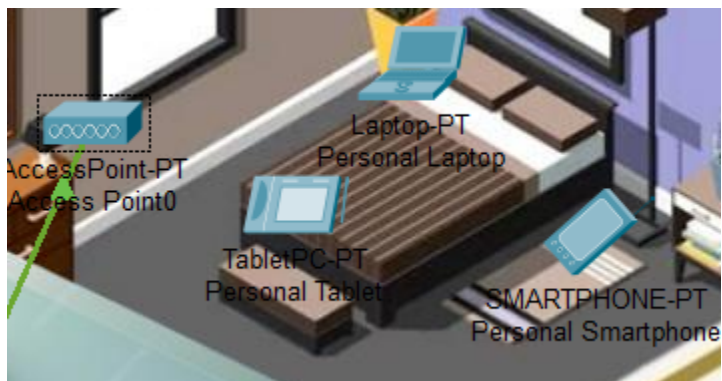
Wireless Connections

1. Access Point 1 (Personal Devices)



- **VLAN:** 20 (Personal Devices)
- **Connection:** FastEthernet 0/11 (Switch)
- **Purpose:** Provides wireless connectivity to personal devices.
- **Connected Devices:**
 - **Tablet**
 - **Smartphone**
 - **Laptop**
- **Location:** In the bedroom

2. Access Point 2 (Guest Devices)



- **VLAN:** 30 (Guest Devices)

- **Connection:** FastEthernet 0/22 (Switch)
- **Purpose:** Provides wireless connectivity to guest devices.
- **Connected Devices:**
 - **Guest Laptop**
 - **Guest Smartphone**
 - **Guest Tablet**
- **Location:** In the living room

Additional Notes:

- **Device Placement:**
 - The table hosts the DSL modem, PC, printer, and phone, which are all connected via Ethernet cables. The table's proximity to the switch and router is essential to minimize cable lengths and reduce signal interference.
- **Wireless Device Distribution:**
 - Devices connected to the Access Points are strategically positioned around the home to ensure optimal wireless coverage. Access Point 1, located in the bedroom, caters to personal devices. This placement provides the best coverage for the client, who frequently uses personal devices in this area. Access Point 2, placed in the living room, is reserved for guest devices. This location ensures that guests have fast and reliable internet connectivity, keeping traffic segmented and secure.

This low-level documentation details the carefully planned setup of a home office network using Cisco Packet Tracer. The configuration is designed to optimize both wired and wireless connectivity within a small-scale environment, ensuring efficient operation and ease of use.

Key components, including the routers, switch, access points, and end devices, have been strategically placed to minimize cable clutter and enhance network performance. The use of VLANs effectively segments traffic between work, personal, and guest devices, providing both security and organization within the network.

The placement of Access Points in critical areas of the home—such as the bedroom for personal devices and the living room for guest devices—ensures optimal wireless coverage and a seamless user experience.

The home office network is equipped to handle daily tasks efficiently, providing a reliable, secure, and scalable networking solution tailored to the client's needs.